

CSMFAB000000000026 V2.0

Aegis Embark: “Vibra-Volt” Piezoelectric Pedal Cover

Version 2.0 — Revised & Expanded | June 2026

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Δ Change Log V1.0 → V2.0

- Updated piezoelectric material to $\text{KNbO}_3\text{-BaTiO}_3$ (lead-free, $T_c = 310^\circ\text{C}$)
 - Added $\text{ZrB}_2\text{-SiC}$ ceramic pedal tip insert with updated microhardness data (22 GPa)
 - New: FBG (Fiber Bragg Grating) optical bio-feedback replacing metallic strain gauges
 - Updated energy harvest calculation: 10 mW average at 70 kg driver, aggressive driving
 - Solar Cycle 25 SSN ~137 context — EV pedal hardware as GIC conductive pathway
 - Added EV platform compatibility matrix (Tesla, Rivian, Hyundai, Ford)
 - V2.0: YInMn Blue + 2 mol% Ta_2O_5 glaze for enhanced arc-flash protection
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1. The Conductive Pedal Problem

Every standard automotive accelerator, brake, and clutch pedal is machined from aluminum or steel — directly connected to the vehicle chassis. During a Carrington-class GIC event, the induced voltage across the vehicle body can reach 50–200 V. The driver’s foot, pressing a metal pedal, completes a circuit through the body to any grounded surface (seat frame, floor). At 100 V across ~5 k Ω body resistance: $I = 20\text{ mA}$ — above the 10 mA ventricular fibrillation threshold for sensitive individuals.

The Vibra-Volt Pedal Cover eliminates this hazard while harvesting the mechanical energy of foot pressure into a deployable power reserve. It is simultaneously a GIC insulator, a vibration energy harvester, and a biofeedback platform.

2. Material Architecture

2.1 Outer Shell: $\text{ZrB}_2\text{-SiC}$ Ceramic Tip Insert

The load-bearing tip contacting the driver’s shoe sole is a sintered $\text{ZrB}_2\text{-SiC}$ ceramic tile: - **Hardness:** 22 GPa Vickers (2025 measurement on flash-sintered samples) - **Resistivity:** $> 10^{10}\ \Omega\cdot\text{m}$ (below percolation threshold at 12 vol% ZrB_2) - **Melting point:** $> 3200^\circ\text{C}$ — thermally inert under any automotive scenario - **Coating:** YInMn Blue + 2 mol% Ta_2O_5 glaze — NIR rejection + arc repulsion

2.2 Harvesting Layer: $\text{KNbO}_3\text{-BaTiO}_3$ Stack

32-element stack of 0.5 mm thick $\text{KNbO}_3\text{-BaTiO}_3$ bimorph discs: - $d_{33} = 285\text{ pC/N}$ - Output at 200 N pedal force (moderate braking), 2 Hz actuation: $P = 8\text{--}12\text{ mW}$ - Connected to a

supercapacitor buffer (10 F, 5 V) via charge pump IC

Energy budget: $10 \text{ mW} \times 3600 \text{ s/hr driving} = 36 \text{ J/hr}$ — sufficient to power the Schumann entrainment module in the Grounding-Sole continuously.

2.3 Base Shell: Basalt Fiber Reinforced Polymer

- VARTM-molded BFRP snap-fit base
- Universal fitment clips: compatible with OEM pedal boxes (adjustable 40–80 mm width)
- Zero metal fasteners — ceramic retention pins only

2.4 Bio-Feedback: Fiber Bragg Grating Sensors (V2.0)

Two FBG sensors embedded in the BFRP shell measure pedal force and rate: - Wavelength shift: $1 \text{ pm}/\mu\epsilon$ strain sensitivity - No metallic sensing elements — fully GIC-immune optical measurement - Data transmitted via optical fiber to dash display unit (PMMA plastic optical fiber, not glass)

3. Vibration Signature Analysis

3.1 EV-Specific Pedal Vibration Profile

Electric vehicles transmit distinct vibration signatures through pedals not present in ICE vehicles: - **Regen braking onset (0.5–3 Hz):** Torque reversal transients, felt as micro-jerk - **Motor torque ripple (60–180 Hz):** Fundamental inverter switching harmonics - **Battery cooling pump (8–25 Hz):** Acoustic resonance transmitted via floor

The Vibra-Volt attenuates sub-20 Hz signals passively through the silicone elastomer compliance layer ($E = 0.5 \text{ MPa}$, $\eta = 0.3$ loss factor), achieving $\approx 12 \text{ dB}$ at 10 Hz.

At 60–180 Hz motor harmonics: the harvesting stack's mechanical resonance is tuned to 120 Hz (center of primary motor harmonic band), maximizing energy extraction while simultaneously damping transmission to the foot.

4. EV Platform Compatibility Matrix (V2.0)

Platform	Pedal Box Width	Clip Size	Harvest Power	GIC Risk Level
Tesla Model 3/Y	52 mm	Size B	9 mW	Medium (Al chassis)
Tesla Model S/X	58 mm	Size C	11 mW	Medium-High
Rivian R1T/R1S	65 mm	Size D	12 mW	High (large frame)

Platform	Pedal Box Width	Clip Size	Harvest Power	GIC Risk Level
Hyundai IONIQ 5/6	50 mm	Size B	8 mW	Medium
Ford Mach-E	55 mm	Size C	9 mW	Medium
Ford F-150 Lightning	62 mm	Size D	13 mW	High (large truck)

5. Phoenix Protocol

- ZrB₂-SiC tip: crush → re-SPS (94% property retention)
- KNbO₃-BaTiO₃ stack: HNO₃ dissolution → hydroxide co-precipitation → 97% elemental recovery
- BFRP shell: thermal solvolysis at 350°C → 95% basalt fiber recovery (Elium® thermoplastic matrix)
- FBG sensors: PMMA fiber is fully recyclable thermoplastic

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